

Outdoor Air pollution

Outdoor air pollution-sources

Air pollution is the presence of chemicals in the atmosphere in concentrations high enough to harm organisms, ecosystems, or human-made materials, or to alter climate.

Natural sources include wind-blown dust, pollutants from wildfires and volcanic eruptions, and volatile organic chemicals released by some plants.

Most **human inputs** of outdoor air pollutants occur in industrialized and urban areas with their higher concentrations of people, cars, and factories.

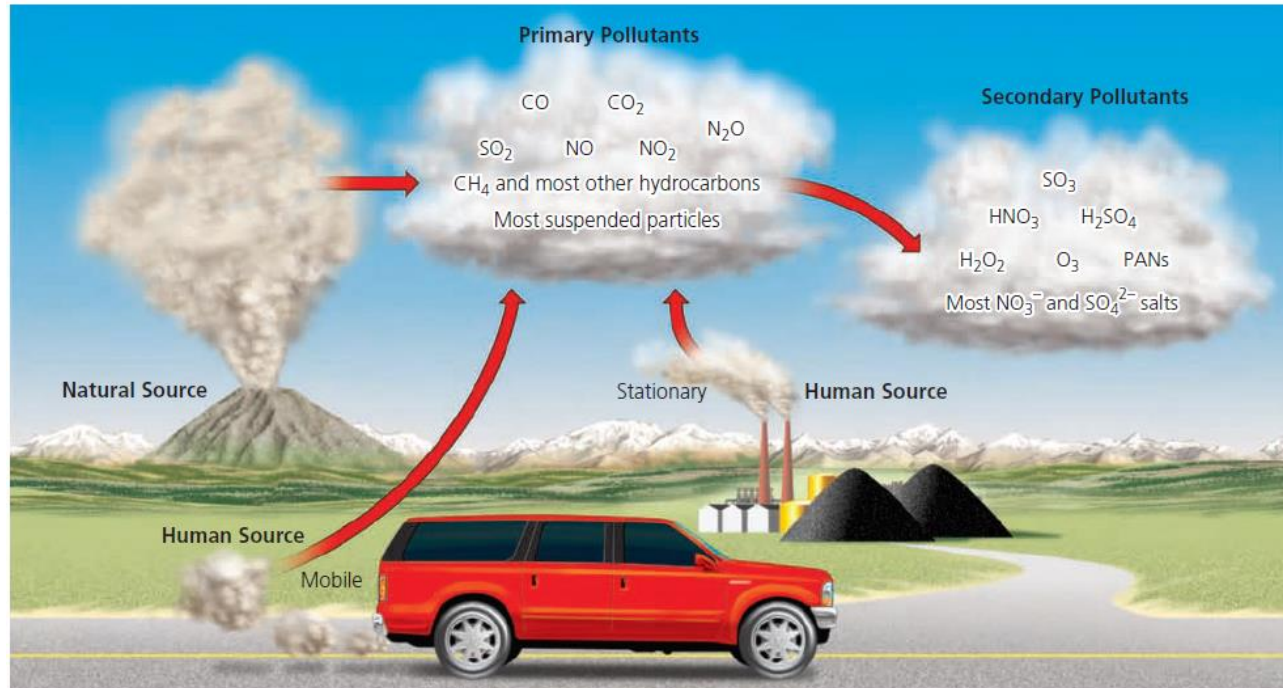
These pollutants are generated mostly by the burning of fossil fuels in power plants and industrial facilities (stationary sources) and in motor vehicles (mobile sources).

Outdoor air pollution-classification

Scientists classify outdoor air pollutants into two categories:

Primary pollutants are chemicals or substances emitted directly into the air from natural processes and human activities at concentrations high enough to cause harm.

secondary pollutants are formed when primary pollutants react with one another and with other natural components of air to form new harmful chemicals,



Major air pollutants

Carbon oxides: *Carbon monoxide (CO)*

Carbon monoxide (CO) is a colorless, odorless, and highly toxic gas that forms during the incomplete combustion of carbon-containing materials

Sources: motor vehicle exhaust,
burning of forests and grasslands,
smokestacks of fossil fuel-burning power plants and industries,
tobacco smoke,
inefficient stoves used for cooking.

- effects:
- 1) Carbon monoxide can combine with hemoglobin in red blood cells, which prevents the normal binding of oxygen with hemoglobin molecules. This in turn reduces the ability of blood to transport oxygen to body cells and tissues.
 - 2) Long-term exposure can trigger heart attacks and aggravate lung diseases such as asthma and emphysema.
 - 3) At high levels, CO can cause headache, nausea, drowsiness, confusion, collapse, coma, and death.

Major air pollutants

Carbon oxides: *Carbon dioxide (CO₂)*

Carbon dioxide (CO₂) is a colorless, odorless gas. About 93% of the CO₂ in the atmosphere is the result of the natural carbon cycle

Sources: the burning of fossil fuels,
the clearing of CO₂-absorbing forests and grasslands.

effects: 1) atmospheric warming and projected climate change

2) Climate change can cause the reduction of food supplies in some areas, while causing water shortages, prolonged drought, or excessive flooding in other areas.

Major air pollutants

Nitrogen oxides and nitric acid.

Sources:

- Nitric oxide (NO) forms when nitrogen and oxygen gas react under high-combustion temperatures in automobile engines and coal-burning power and industrial plants.
- In the air, NO reacts with oxygen to form nitrogen dioxide (NO₂), a reddish-brown gas. Collectively, NO and NO₂ are called nitrogen oxides (NO_x).
- Nitrous oxide (N₂O) is emitted from fertilizers and animal wastes and is produced by the burning of fossil fuels.

effects:

- 1) Both NO and NO₂ play a role in the formation of photochemical smog
- 2) Nitrous oxide (N₂O), a greenhouse gas
- 3) Some of the NO₂ reacts with water vapor in the air to form nitric acid (HNO₃) and nitrate salts (NO₃⁻), components of harmful acid deposition,

Major air pollutants

Sulfur dioxide and sulfuric acid

Sulfur dioxide (SO₂) is a colorless gas with an irritating odor

Sources:

- About one third of the SO₂ in the atmosphere comes from natural sources as part of the sulfur cycle
- The other two-thirds come from combustion of sulfur-containing coal in power and industrial plants, oil refining and smelting of sulfide ores.

effects:

- 1) Sulfur dioxide, sulfuric acid droplets, and sulfate particles reduce visibility and aggravate breathing problems.
- 2) They can also damage crops, trees, soils, and aquatic life in lakes, and they corrode metals and damage paint, paper, leather, and stone on buildings and statues
- 3) In the atmosphere, SO₂ can be converted to aerosols, which consist of microscopic suspended droplets of sulfuric acid (H₂SO₄) and suspended particles of sulfate(SO₄²⁻) salts that return to the earth as a component of acid deposition

Major air pollutants

Particulates

Suspended particulate matter (SPM) consists of a variety of solid particles and liquid droplets that are small and light enough to remain suspended in the air for long periods. The EPA classifies particles as fine, or PM-10 (with diameters less than 10 micrometers), and ultrafine, or PM-2.5 (with diameters less than 2.5 micrometers).

Sources:

- natural sources such as dust, wild fires, and sea salt
- human sources such as coal-burning power and industrial plants, motor vehicles, and road construction.

effects:

- 1) These particles can irritate the nose and throat, damage the lungs, aggravate asthma and bronchitis, and shorten life.
- 2) Toxic particulates such as lead, cadmium, and polychlorinated biphenyls can cause genetic mutations, reproductive problems, and cancer
- 3) Particulates also reduce visibility, corrode metals, and discolor clothes and paints.

Major air pollutants

Ozone

- A colorless and highly reactive gas, is a major ingredient of photochemical smog.
- Ozone in the troposphere near ground level is often referred to as "bad" ozone, while we view ozone in the stratosphere as "good" ozone because it protects us from harmful UV radiation.

Sources:

human activities are decreasing the amount of beneficial ozone in the stratosphere and increasing the amount of harmful ozone in the troposphere especially in some urban areas.

Effects:

It can cause coughing and breathing problems, aggravate lung and heart diseases, reduce resistance to colds and pneumonia, and irritate the eyes, nose, and throat. It also damages plants, rubber in tires, fabrics, and paints.

Major air pollutants

Volatile organic compounds (VOCs).

- Organic compounds that exist as gases in the atmosphere or that evaporate from sources on earth into the atmosphere are called volatile organic compounds (VOCs). Example: Methane (CH_4)
- Other VOCs are liquids that can evaporate into the atmosphere. Examples : benzene and other liquids used as industrial solvents, dry-cleaning fluids, and various components of gasoline, plastics

Sources:

- About a third of global methane emissions comes from natural sources, mostly plants, wetlands, and termites.
- rice paddies, landfills, oil and natural gas wells, and cows (mostly from their belching).

Effects:

methane (CH_4), a greenhouse gas that is 20 times more effective per molecule than CO_2 is at warming the atmosphere through the greenhouse effect

Air pollution problems

Smog

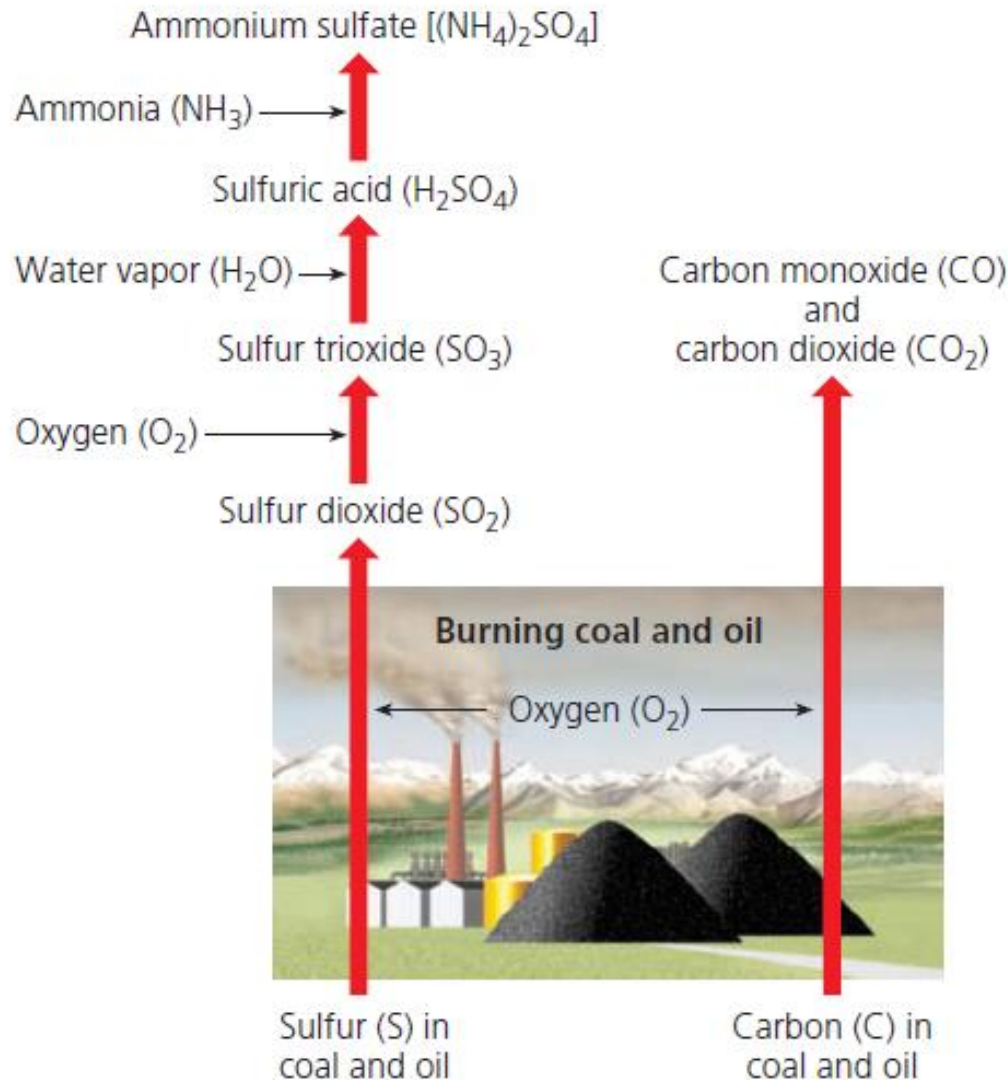
Smog which occurs mainly because of air pollution can also be defined as a mixture of various gases with dust and water vapor. Smog also refers to hazy air that makes breathing difficult.

- Industrial smog
- Photochemical Smog

Air pollution problems

Industrial smog

- Industrial smog consists mostly an unhealthy mix of sulfur dioxide, suspended droplets of sulfuric acid, and a variety of suspended solid particles.



Air pollution problems

Industrial smog

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Carbon (C) in Burning
Coal and oil → carbon monoxide (CO) and carbon dioxide (CO₂) and **soot** particles

Sulfur (S) in Burning
Coal and oil → Sulfur dioxide (SO₂) Oxygen → Sulfur trioxide (SO₃)

↓ Water vapor

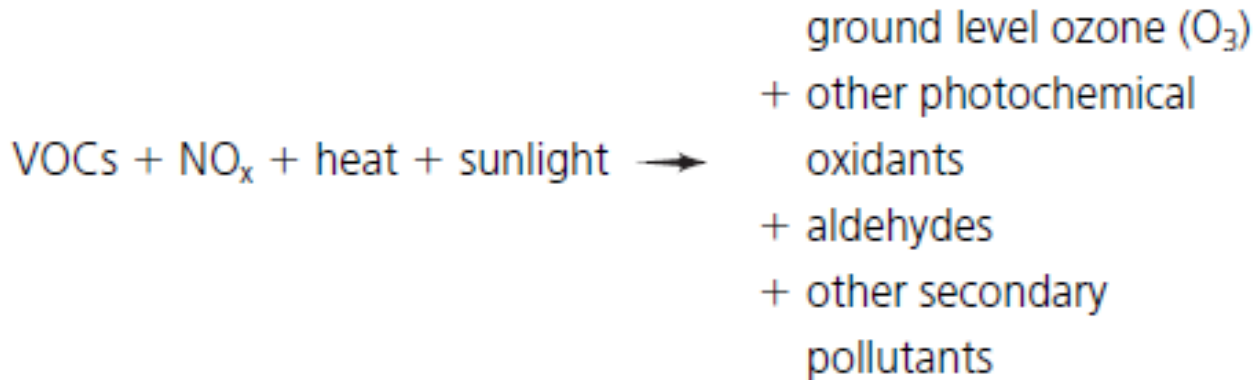
Ammonium sulfate (NH₄)₂SO₄ ← Ammonia → Sulfuric acid (H₂SO₄)

The suspended particles of **ammonium sulfate** salts and **soot** give the resulting smog a gray color, which is why it is sometimes called **gray-air smog**

Air pollution problems

Photochemical smog

Photochemical smog is a mixture of primary and secondary pollutants formed under the influence of UV radiation from the sun.



The photochemical smog is a mixture of ozone, nitric acid, aldehydes, peroxyacyl nitrates (PANs), and other secondary pollutants.

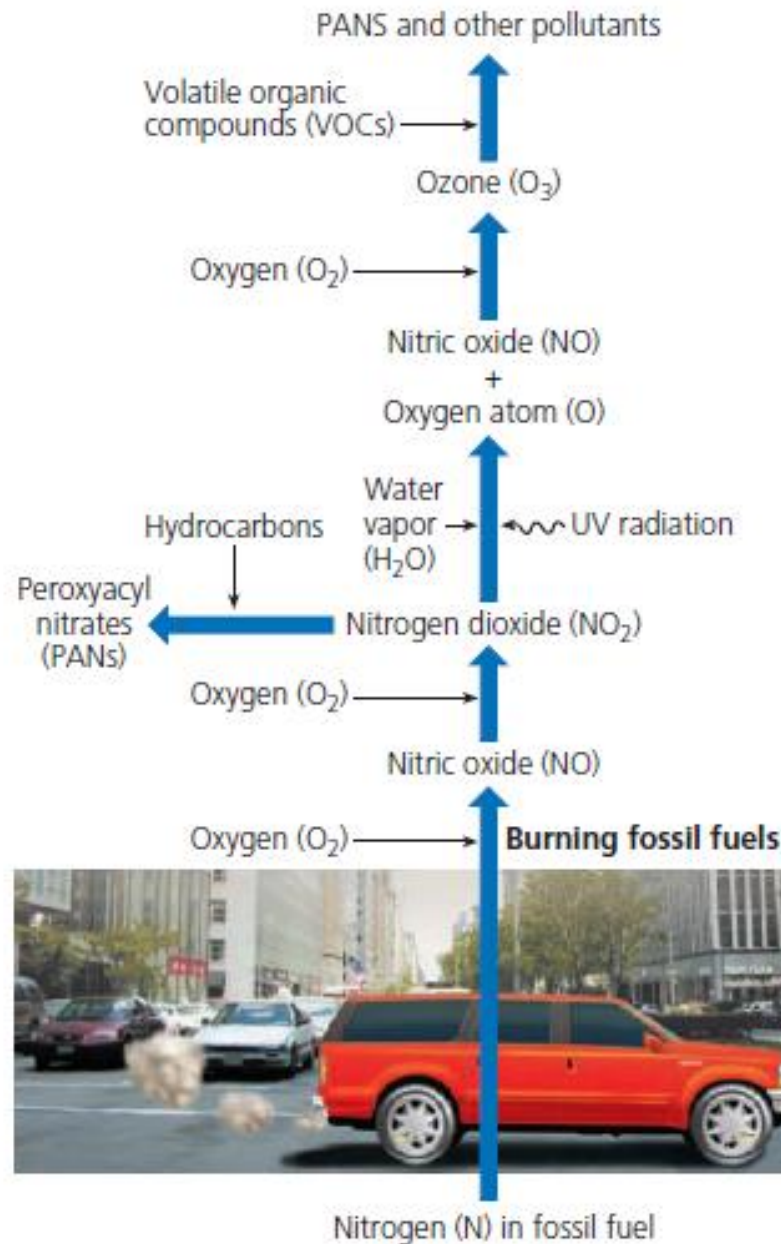
Air pollution problems

Photochemical smog formation

- The formation of photochemical smog begins when exhaust from morning commuter traffic releases large amounts of NO and VOCs into the air over a city.
- The NO is converted to reddish-brown NO₂, explaining why photochemical smog is sometimes called **brown-air smog**.
- When exposed to ultraviolet radiation from the sun, some of the NO₂ reacts in complex ways with VOCs released by certain trees (such as some oak species, sweet gums, and poplars), motor vehicles, and businesses (such as bakeries and dry cleaners).

Air pollution problems

Photochemical smog



The already poor air quality in urban areas of many less-developed countries is worsening as the number of motor vehicles rises.

Many of these vehicles are 10 or more years old, have no pollution control devices, and burn leaded gasoline.

Acid deposition

Acid deposition

Acid rain, or acid deposition, is a broad term that includes any form of precipitation with acidic components, such as sulfuric or nitric acid that fall to the ground from the atmosphere in wet or dry forms. This can include rain, snow, fog, hail or even dust that is acidic.

What Causes Acid Rain?

Acid rain results when sulfur dioxide (SO_2) and nitrogen oxides (NO_x) are emitted into the atmosphere and transported by wind and air currents.

The SO_2 and NO_x react with water, oxygen and other chemicals to form sulfuric and nitric acids. These then mix with water and other materials before falling to the ground.

Natural causes

Human activities

Acid deposition

Natural causes:

Lightning strikes : formation of nitrogen oxides

volcanic eruptions: release of sulfur dioxide

Human activities:

- Burning of fossil fuels to generate electricity. Two thirds of SO_2 and one fourth of NO_x in the atmosphere come from electric power generators.
- Vehicles and heavy equipment.
- Manufacturing, oil refineries and other industries.

Winds can blow SO_2 and NO_x over long distances and across borders making acid rain a problem for everyone and not just those who live close to these sources.

Acid deposition

Forms of acid deposition

The acidic substances remain in the atmosphere for 2-14 days, depending mostly on prevailing winds, precipitation, and other weather patterns. During this period, they descend to the earth's surface in two forms:

Wet deposition and Dry deposition

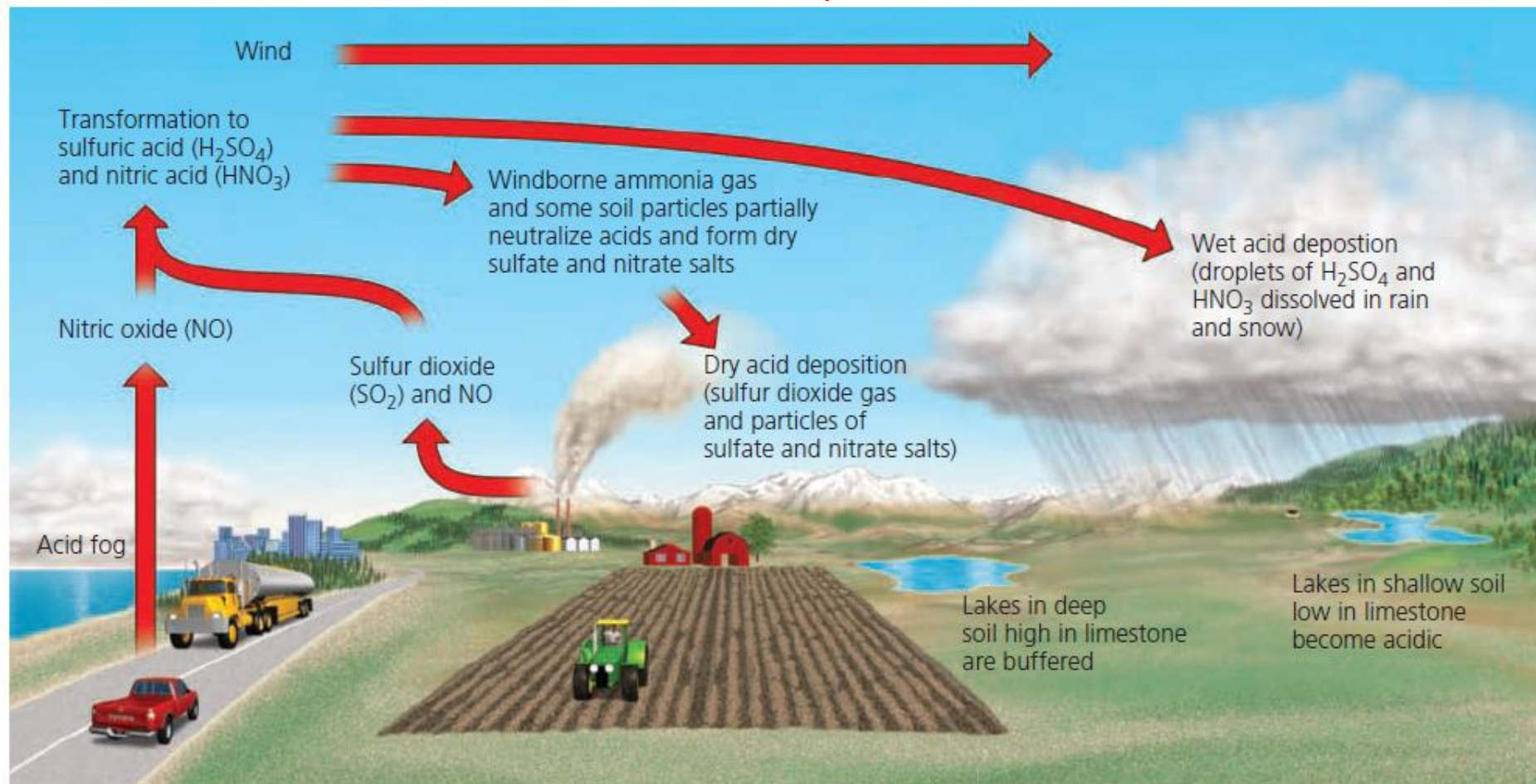
Wet deposition : wet deposition, consisting of acidic rain, snow, fog, and cloud vapor with a pH of less than 5.6, and most wet deposition takes place within 4-14 days of emission in more distant downwind areas

Dry deposition : Acidic particles and gases can also deposit from the atmosphere in the absence of moisture as dry deposition.

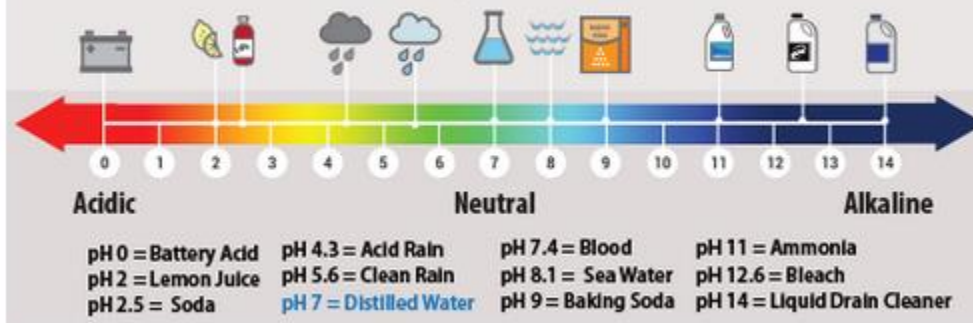
The acidic particles and gases may deposit to surfaces (water bodies, vegetation, buildings) quickly or may react during atmospheric transport to form larger particles that can be harmful to human health.

Most dry deposition occurs within 2-3 days of emission, fairly near the industrial sources,

Acid deposition



The pH Scale



Acid deposition

Harmful effects

Effects on aquatic ecosystem:

- Acid deposition harms aquatic ecosystems. Most fish cannot survive in water with a pH less than 4.5
- Acid deposition can also release aluminum ions (Al^{3+}), which are attached to minerals in some soils bordering lakes. These ions asphyxiate many kinds of fish by stimulating excessive mucus formation, which clogs their gills.
- Aerosols in the clouds are pulled into thunderstorms that then dump acid rain into the Indian and Pacific Oceans, possibly harming marine life and ecosystems.

Acid deposition-Harmful effects

Effects on crop production:

- Acid deposition (often along with other air pollutants such as ozone) can harm crops, especially when the soil pH is below 5.1. Low pH reduces plant productivity and the ability of soils to buffer or neutralize acidic inputs.
- An estimated 30% of China's cropland suffers from excess acidity.

Effects on forests:

- Acid rain can leach essential plant nutrients such as calcium and magnesium from forest soils.
- Acid rain can release ions of aluminum, lead, cadmium, and mercury to the soil where they can damage tree roots.
- Mountaintop forests are the terrestrial areas hardest hit by acid deposition. These areas tend to have thin soils without much buffering capacity

How to Reduce Acid Deposition

- The best solutions are *preventive approaches* that reduce or eliminate emissions of sulfur dioxide, nitrogen oxides, and particulates.

Prevention

- Reduce coal use
- Burn low-sulfur coal
- Increase use of natural gas and renewable energy resources
- Remove SO_2 particulates and NO_x from smokestack gases and remove NO_x from motor vehicular exhaust
- Tax emissions of SO_2

Cleanup

- Add lime to neutralize acidified lakes
- Add phosphate fertilizer to neutralize acidified lakes

Indoor Air pollution

Indoor Air pollution

According to the WHO, indoor air pollution is the world's most serious air pollution problem, especially for **poor people**. This is a glaring example of the connections between poverty, environmental quality, and human health.

the indoor burning of wood, charcoal, dung, crop residues, coal, and other fuels in open fires or in unvented or poorly vented stoves exposes people to dangerous levels of particulate air pollution.



Indoor Air pollution

- Indoor air pollution is also a serious problem in developed areas of all countries, mostly because of chemicals used in building materials and products.

Chloroform

Source: Chlorine-treated water in hot showers

Possible threat: Cancer

1,1,1-Trichloroethane

Source: Aerosol sprays

Threat: Dizziness, irregular breathing

Nitrogen oxides

Source: Unvented gas stoves and kerosene heaters, woodstoves

Threat: Irritated lungs, children's colds, headaches

Asbestos

Source: Pipe insulation, vinyl ceiling and floor tiles

Threat: Lung disease, lung cancer

Particulates

Source: Pollen, pet dander, dust mites, cooking smoke particles

Threat: Irritated lungs, asthma attacks, itchy eyes, runny nose, lung disease

Para-dichlorobenzene

Source: Air fresheners, mothball crystals

Threat: Cancer

Tetrachloroethylene

Source: Dry-cleaning fluid fumes on clothes

Threat: Nerve disorders, damage to liver and kidneys, possible cancer

Indoor Air pollution

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Formaldehyde

Source: Furniture stuffing, paneling, particleboard, foam insulation

Threat: Irritation of eyes, throat, skin, and lungs; nausea; dizziness

Radon-222

Source: Radioactive soil and rock surrounding foundation, water supply

Threat: Lung cancer

Carbon monoxide

Source: Faulty furnaces, unvented gas stoves and kerosene heaters, woodstoves

Threat: Headaches, drowsiness, irregular heartbeat, death

Styrene

Source: Carpets, plastic products

Threat: Kidney and liver damage

Tobacco smoke

Source: Cigarettes

Threat: Lung cancer, respiratory ailments, heart disease

Benzo- α -pyrene

Source: Tobacco smoke, woodstoves

Threat: Lung cancer

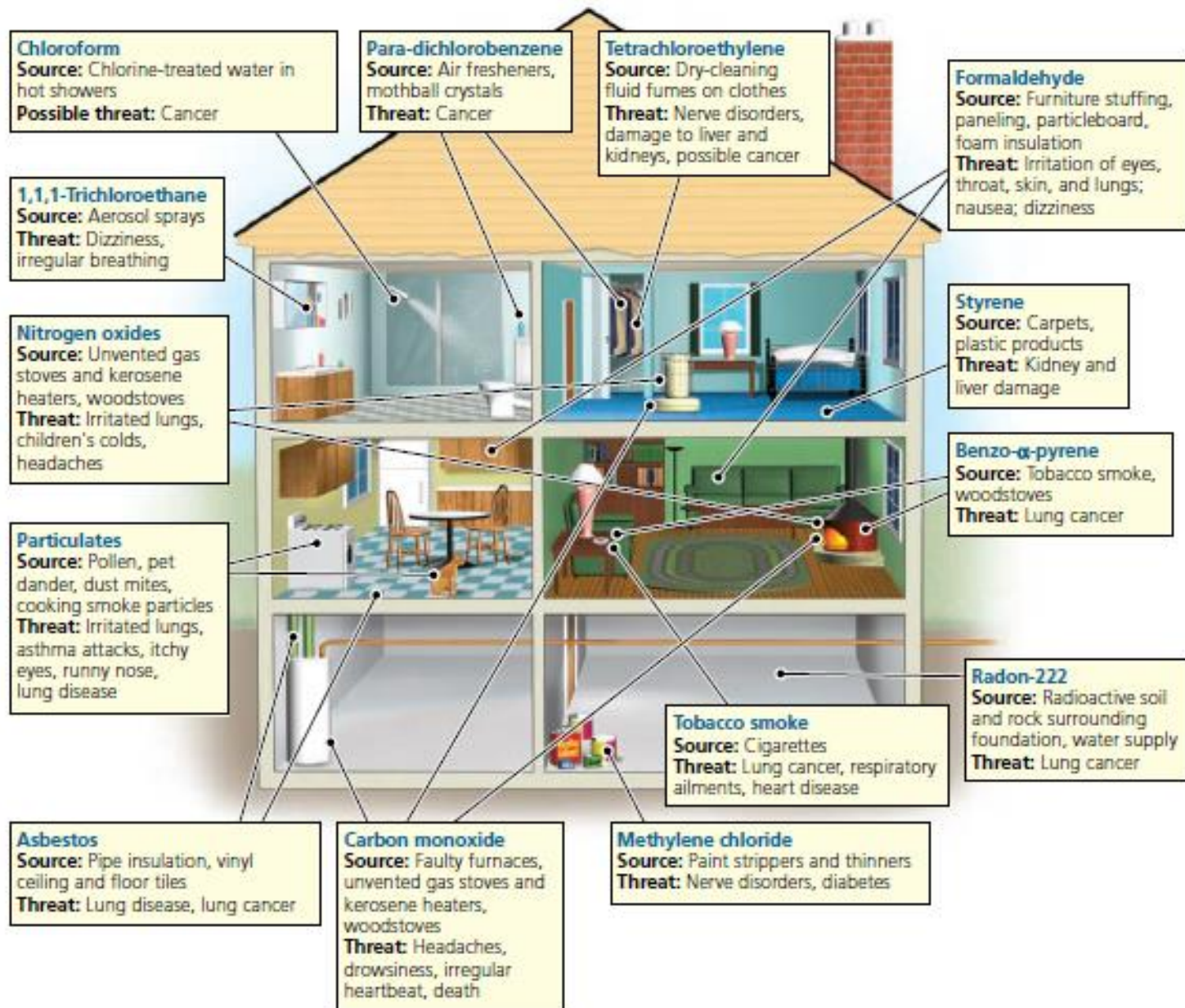
Methylene chloride

Source: Paint strippers and thinners

Threat: Nerve disorders, diabetes

Indoor Air pollution

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Indoor Air pollution

Four most dangerous indoor air pollutants:

According to the public health officials, the four most dangerous indoor air pollutants in **more developed countries are**

- Tobacco smoke
- Formaldehyde emitted from many building materials and various household products.
- radioactive radon-222 gas, which can seep into houses from underground rock deposits
- very small (ultrafine) particles of various substances in emissions from motor vehicles, coal burning and industrial power plants, wood burning and forest and grass fires.

Indoor Air pollution

Pollutant: Formaldehyde

The chemical that causes most people in more developed countries difficulty is **formaldehyde** (CH_2O , a colorless, extremely irritating chemical).

Source: Building materials (such as plywood, high-gloss wood used on floors and for cabinets), furniture, drapes, adhesives in carpeting and wallpaper, urethane- formaldehyde foam insulation, fingernail hardener, and wrinkle-free coatings on permanent-press clothing.

Effect:

According to EPA and the American Lung Association, 20-40 million Americans suffer from chronic breathing problems, dizziness, rashes, headaches, sore throats, sinus and eye irritation, skin irritation, wheezing, and nausea caused by daily exposure to low levels of formaldehyde emitted from common household materials.

Indoor Air pollution

Pollutant: Radioactive Radon Gas

Radon-222 is a colorless, odorless, radioactive gas that is produced by the natural radioactive decay of uranium-238, small amounts of which are contained in most rocks and soils.

Source:

Uranium-238 isotope is much more concentrated in underground deposits of minerals such as uranium, phosphate, granite.

In buildings above such deposits, radon gas can enter through cracks in a foundation's slab and walls, openings around sump pumps and drains, and hollow concrete blocks. Once inside, it can build up to high levels, especially in unventilated lower levels of homes and buildings.

Effect:

Radon-222 gas quickly decays into solid particles of other radioactive elements such as polonium-210, which if inhaled, expose lung tissue to large amounts of ionizing radiation from alpha particles. This exposure can damage lung tissue and lead to lung cancer over the course of a 70-year lifetime.

Indoor Air pollution

Facts about indoor air pollution:

EPA studies have revealed some alarming facts about indoor air pollution.

- First, levels of 11 common pollutants generally are 2 to 5 times higher inside homes and commercial buildings than they are outdoors, and in some cases they are as much as 100 times higher
- Second, pollution levels inside cars in traffic-clogged urban areas can be up to 18 times higher than outside levels.
- Third, the health risks from exposure to such chemicals are magnified because most people in developed urban areas spend 70-98% of their time indoors or inside vehicles.

How to improve air quality?

How to improve air quality

Laws and regulations can reduce outdoor air pollution

- There is an establish air quality standards for six major outdoor pollutants—carbon monoxide (CO), nitrogen dioxide (NO₂), sulfur dioxide (SO₂), suspended particulate matter (SPM, smaller than PM-10), ozone (O₃), and lead (Pb).

One limit, called a **primary standard**, is set to protect human health. Another limit, called a **secondary standard**, is intended to prevent environmental and property damage.

Each standard specifies the maximum allowable level for a pollutant, averaged over a specific period.

- Established national emission standards for more than 188 hazardous air pollutants (HAPs) that may cause serious health and ecological effects

How to improve air quality

Laws and regulations can reduce outdoor air pollution

- Putting much greater emphasis on preventing air pollution. The prevention is always better choice. Example: In the United States, the air pollutant with the largest drop in its atmospheric level was lead, which was largely banned in gasoline
- Refineries, power plants, mines, chemical manufacturers, and factories are required report their releases and waste management methods for toxic chemicals.
- Improving fuel efficiency standards for motor vehicles
- More strictly regulating emissions from motorcycles and two-cycle gasoline engines.
- Setting much stricter air pollution regulations for airports and oceangoing ships
- Greatly increasing and modernizing the monitoring of air pollutants

How to improve air quality

Reducing indoor air pollution

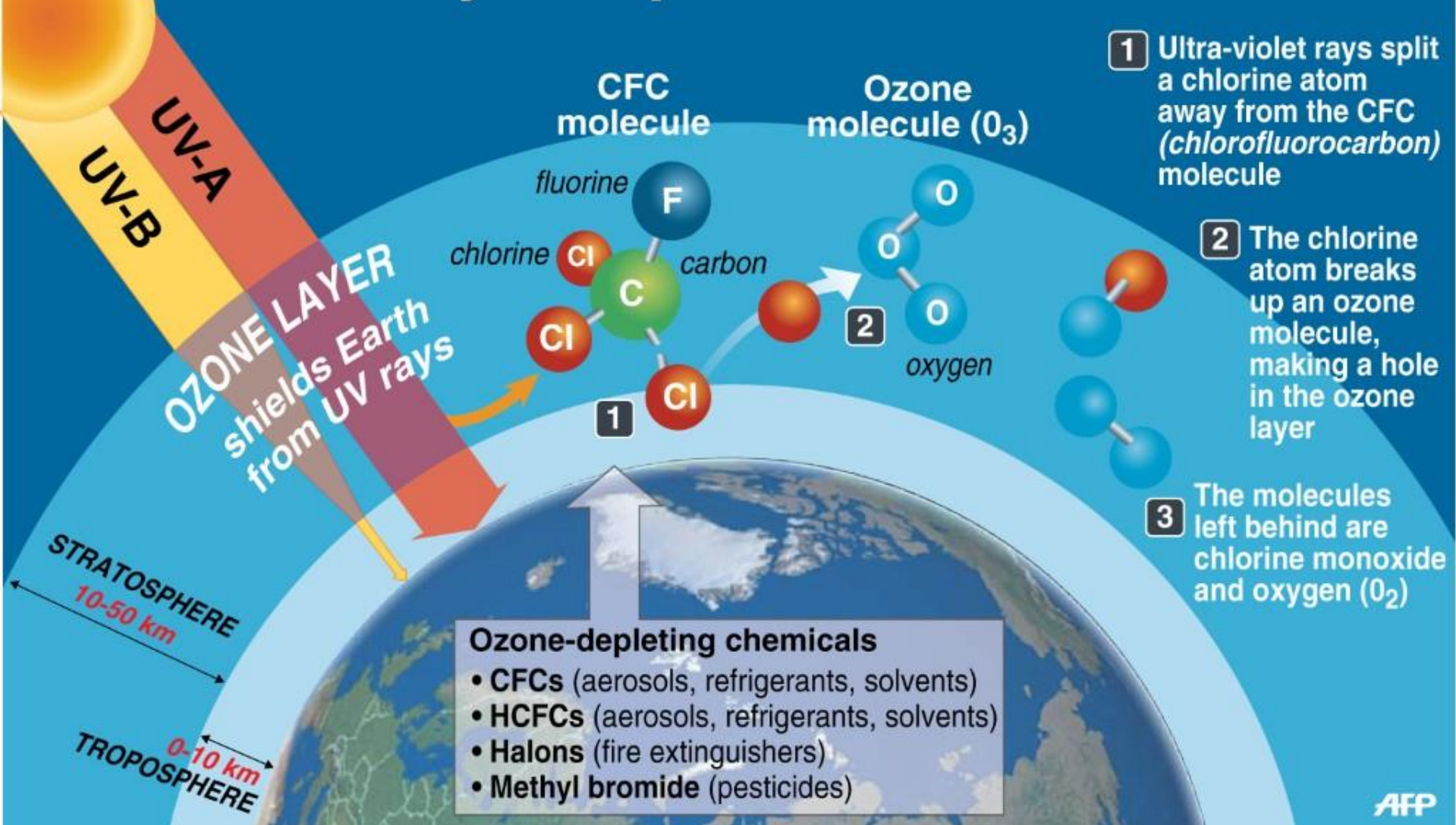
Prevention

- Ban indoor smoking
- Set stricter formaldehyde emissions standards for carpet, furniture, and building materials
- Prevent radon infiltration
- Use less polluting cleaning agents, paints, and other products

Cleanup or dilution

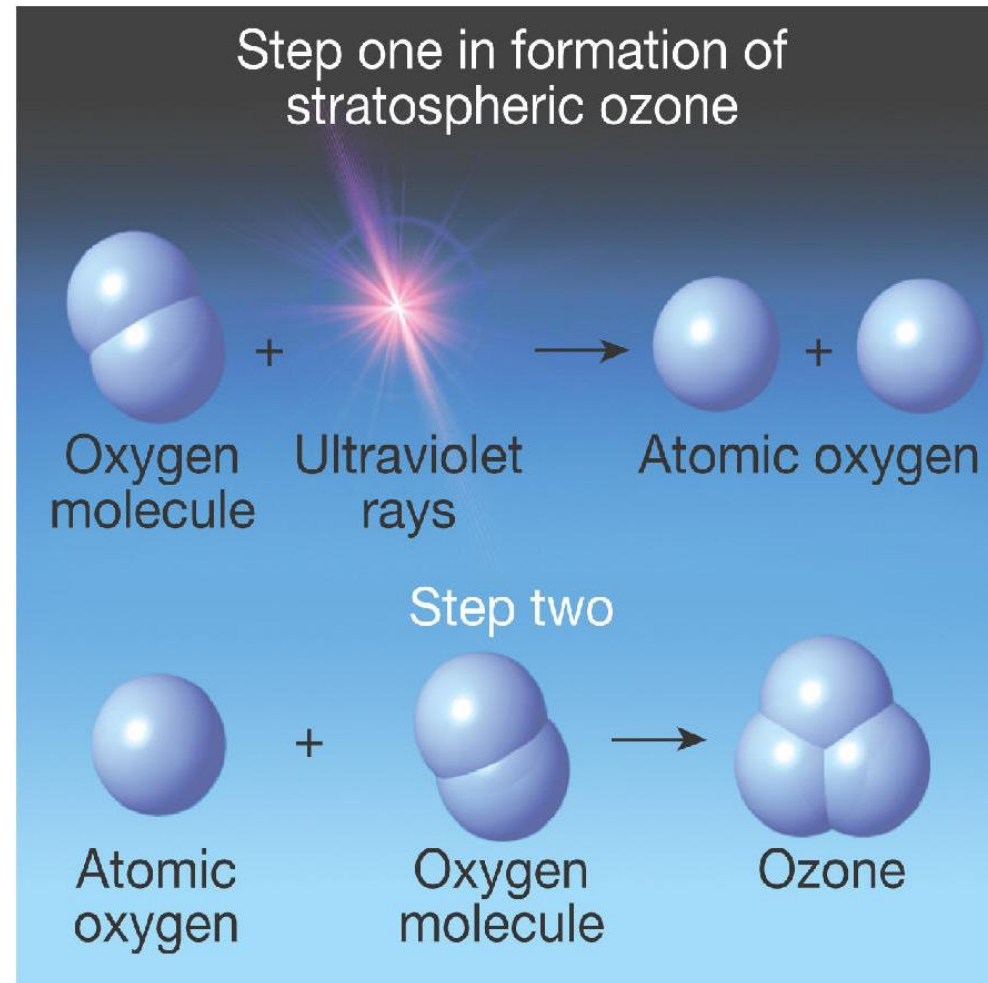
- Use adjustable fresh air vents for work spaces
- Circulate air more frequently
- Circulate a building's air through rooftop greenhouses
- Use efficient venting systems for wood-burning stoves

Ozone layer depletion



Formation and Breakdown of Stratospheric Ozone

- Stratospheric ozone forms and breaks down naturally in a ***closed-loop cycle***
- UV-C radiation breaks the bond, that holds the oxygen molecule together, leaving ***two free oxygen atoms***
- The free oxygen atoms ***react with oxygen molecule*** to form ozone
- Ozone is broken down into oxygen molecule when it absorbs both UV-C and UV-B light. This free oxygen again may react with O_2 to form ozone. ***Thus ozone is formed and broken down in a cycle.***



Anthropogenic Contribution to Ozone Destruction

- Certain chemical catalysts can break down ozone, the most important one is **chlorine**.
- Major source of chlorine in stratosphere is a class of anthropogenic compounds known as **chlorofluorocarbons (CFCs)**.
- CFCs are the family of organic compounds whose properties make them ideal for use in **refrigeration and air conditioning**, as **propellant in aerosol cans to deliver ingredients such as deodorant and insect repellent**, and as **blowing agents to inject air into foam products like Styrofoam cups and foam insulation**.
- CFCs **does not degrade**, dissolve in water or undergo any significant chemical change, but slowly circulates in the atmosphere.
- However, **in presence of UV radiation** in stratosphere, the chlorine bond in CFC breaks

Depletion of Ozone Layer

- The stratospheric ozone in Antarctica had been *decreasing each year*, from 1979.
- It had also been observed that, in the Antarctica, ozone depletion was seasonal: each year depletion occurred from August through November.
- The depletion caused an area of severely reduced ozone concentrations over most of Antarctica, called as “*Ozone hole*”
- Decreased stratospheric ozone has increased the amount of UV-B radiation that reaches the earth's surface